



Flame properties and maximum ceiling smoke temperature in tunnel fires with two asymmetric fire sources under natural ventilation

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ABSTRACT

Multiple fire source fires in a tunnel may cause serious fire consequences. In the present study, the flame properties and maximum ceiling smoke temperature for two asymmetric fire sources in the tunnel with natural ventilation are studied in detail. A large number of model-scale fire experiments are performed, considering energy release rate ratio and fire source spacing. The results show that the smaller fire has a larger flame inclination angle and the flame inclination angle of the smaller fire increases with energy release rate ratio. In contrast, the flame tilt angle of the larger fire becomes slightly smaller with the energy release rate ratio. When mean flames merge vertically, the flame height becomes higher when the energy release rate ratio increases. A revised model based on the air entrainment perimeter is proposed to calculate the height of the merged flame. When mean flames cannot merge vertically, a non-dimensional correlation is also developed. With a longer fire source spacing, the maximum ceiling smoke temperature decreases first and then remains unchanged. According to the relative position of the flame tip and tunnel ceiling height, a model is developed to calculate the maximum ceiling smoke temperature. The results of this paper are significant to the fire detection and structure protection for the possibility of two asymmetric fire sources in a tunnel with natural ventilation.

1. Introduction

Fire accidents are one of the most fatal problems affecting traffic safety operations (Ingason et al., 2014). Because of the long-narrow space structure, fire smoke can spread rapidly along tunnel, which is hard to discharge from the tunnel. Long-time exploration of the high-temperature smoke environment poses a significant adverse effect on the stability of the structure. There may cause heavy casualties when a tunnel fire occurs. For instance, 39 people died and 40 vehicles involved in Mount-Blanc tunnel fire in 1999 and the tunnel was closed for 3 years because of the severe tunnel structure damage. In 2014, the Yanhou tunnel fire in China involved 42 vehicles and led to 40 deaths. Considering the grave danger of the tunnel fire, extensive studies have been conducted all over the world, including flame properties (Kurioka et al., 2003), maximum ceiling smoke temperature (Hu et al., 2006; Li et al., 2011; Yao et al., 2020), critical velocity and smoke back-layering

length (Hu et al., 2008; Li et al., 2010; Chow et al., 2015; Weng et al., 2016), and smoke extraction efficiency (Cong et al., 2021; Wang et al., 2021; Jiang et al., 2022).

Because of vehicle collisions, multiple fire sources phenomenon may occur, that is, more than two vehicles may burn at the same time (He et al., 2024b). Different from the burning characteristics of a single fire, more fuel burns at the same time in multi-fire scenarios. This will lead to a larger total energy release rate (Ingason et al., 2014; He et al., 2024a), and the damage to the tunnel structure is even worse. Especially, when the fire source spacing is relatively short, the flames can merge together accompanied by a higher flame height. As a result, a growing number of scholars begin to investigate the multiple fire sources in long-narrow confined space. Ji et al. (Ji et al., 2016) investigated the combustion properties of two symmetrical fires in the naturally ventilated tunnel. In their study, the effect of distance between the burners on burning rates and flame merging behaviors was discussed in detail. Chen et al. (Chang-kun et al., 2016) discussed the influence of portal seal ratios on

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Nomenclature

c_p	specific heat of air at constant pressure (kJ/(kg·K))
D	side length of the burner (m)
D_{eq}	equivalent burner diameter (m)
H_{ef}	effective tunnel height (m)
g	gravitational acceleration (m/s ²)
P	air entrainment boundary (m)
Q_{total}	total energy release rate of two fire sources (kW)
Q_p^*	non-dimensional energy release rate based on air entrainment boundary
Q_D^*	non-dimensional energy release rate based on equivalent diameter
T_∞	temperature of the ambient air (K)
ΔT_{max}	maximum excess gas temperature beneath tunnel ceiling (K)
N	energy release rates ratio
S	fire source spacing from burner side to burner side (m)
S'	fire source spacing from burner center to burner center (m)
S_c	critical flame merging fire source spacing (m)
Z_f	flame height (m)
$Z_{f,S=0}$	flame height with zero fire source spacing (m)
$Z_{f,S}$	flame height driven by a single fire source (m)
Greek symbols	
ρ_a	density of ambient air (kg/m ³)

combustion behaviors of multiple pool fires by model-scale fire tests. Fan and Tang (Fan and Tang, 2017) analyzed the interaction mechanism between the laterally arranged twin fires in the tunnel with longitudinal ventilation. Tang et al. (Tang et al., 2020) studied the critical longitudinal ventilation velocity for smoke control in a tunnel induced by two nearby fires of various distances. Jia et al. (Jia et al., 2021) analyzed mass loss rate through a large number of model-scale fire experiments, considering different distances between the two fire sources. Zhao et al. (Zhao et al., 2022) revealed burning behaviors of the double fires with the same fuel pan size in the tunnel with natural ventilation. Ceiling smoke temperature distribution for two fires was also been studied in detail (Zhang et al., 2020; Ren et al., 2021; He et al., 2023). Recently, Guo et al. (Guo et al., 2023) discussed the fire behaviors of two fires with the same fuel pan size through full-scale fire experiments under different ventilations. Wan et al. (Wan et al., 2022; Wan et al., 2023) researched the flame interaction and extension beneath the ceiling for the two asymmetric fire sources. The flame merging behaviors and smoke backlayering length of two fires in a longitudinally ventilated tunnel were also investigated (Meng, 2023).

However, the research on asymmetric two fire sources with unequal energy release rates is relatively limited. In the actual tunnel fire accident, the fire loads of different vehicles are disparate and the ignition time is often not completely consistent. As a result, asymmetric multiple fire sources may be more common. A deep understanding of the fire properties with two asymmetric fire sources in the long-narrow space is still lacking. Therefore, flame properties and maximum ceiling smoke temperature in tunnel fires with two asymmetric fire sources under natural ventilation were mainly focused on. A large number of model-scale tests are conducted systematically, considering asymmetric ratio and fire source spacing.

2. Experimental procedure

In fire search, the Froude scaling law is often adopted to carry out reduced-scale tunnel tests (Oka and Imazeki, 2014; Yao et al., 2016). In

this study, a 1:12 model-scale tunnel was used to perform the fire tests. The dimensions of the model tunnel are 14 m long, 0.5 m high and 0.4 m wide. To avoid the influence of wind on flame properties during test, this model-scale tunnel is placed in a large laboratory. The ambient pressure in this study is at normal atmospheric pressure with 0.101 MPa. The ambient temperature varies from 20°C to 25°C.

For fire source, the square gas burner was used in this study and its side length is 10 cm. The burner was placed 2 cm above the tunnel floor along the tunnel axis. Propane gas was selected as fuel because of its relatively high combustion efficiency. To precisely adjust the fire size of each burner, two separated digital mass flow controllers (Sevenstar CS200, accuracy: $\pm 1.0\%$, S.P. ($\geq 35\%$ F.S.), $\pm 0.35\%$ F.S. ($< 35\%$ F.S.)) were used to control the fuel mass flow rate of each burner. Here, the energy release rate is estimated using $\dot{Q} = \chi \dot{m}_{fuel} \Delta H_c$. The complete heat of combustion for propane is 46.45 MJ/kg. Based on the previous studies (Gao et al., 2015; Zhang et al., 2016; He et al., 2023), the combustion efficiency for propane could be assumed to be unite. During the experiment, the centreline of the two burners was kept unchanged. In contrast, the longitudinal position of the two burners was adjusted along the centreline of the model tunnel. In this study, the energy release rate ratio varies from 1 to 4. As shown in Fig. 1, fire source spacing is defined as the distance from burner side to burner side and changes from 0 m to 0.9 m. Total 36 tests were carried out, which are shown in Table 1.

To obtain ceiling smoke temperature, numerous K-type thermocouples were placed 2 cm beneath the ceiling along tunnel axis. The diameter of K-type thermocouple is 1 mm. The temperature collection interval is 1 s. The longitudinal distances between thermocouples range from 5 cm to 10 cm. In order to obtain relevant parameters of the flame, a digital camera was placed at the front end of the model tunnel to record the experimental phenomena. The frequency was set as 25 fps.

3. Results and discussion

3.1. Flame inclination

Different from the single fire, the interaction from the adjacent flame can lead to intricate flame properties. Fig. 2 presents representative pictures of the flame shape under different flame merging states. It can be observed that two flames fully merge from the burner bottom when the distance between the two burners is zero. The merged flame is gradually separated from the bottom to the top of the flame with a longer distance between the two burners. The two flames are fully separated when the horizontal projection distance of the two flames is less than the distance between the two burners. However, the separated flames are still inward-tilted. This is because the longitudinal induced wind will be generated in long-narrow space during a fire (He et al., 2021). For a given fire source spacing, when the energy release rate ratio N increases, the two flames are easier to merge. The reason is that flame height and the flame inward deflection distance increase with the energy release rate. Meanwhile, air entrainment intensity increases when N increases, which results in the increased tilt degree and inward deflection distance of the left flame.

For turbulent diffusion flame in fire research, flames exhibit periodic oscillatory fluctuations (Karlsson and Quintiere, 1999). Therefore, 1500 flame pictures (60 s) were processed using image processing methods to obtain flame characteristic parameters. As the mean flame properties are important in fire research, the mean flame inclination angle is discussed in the following text. Fig. 3 shows flame inclination angle versus fire spacing. In the present study, the flame inclination angle is defined as the angle between the vertical centreline of the burner and the connecting line from the mean flame tip to the center of the burner (Mudan, 1987; Wan et al., 2018; Lin et al., 2019). For cases with mean flames merged ($S = 0$ m and $S = 0.5$ m), it is difficult to obtain the flame inclination angle. Therefore, only experimental data for cases with separated mean flames is shown in Fig. 3. The following phenomena

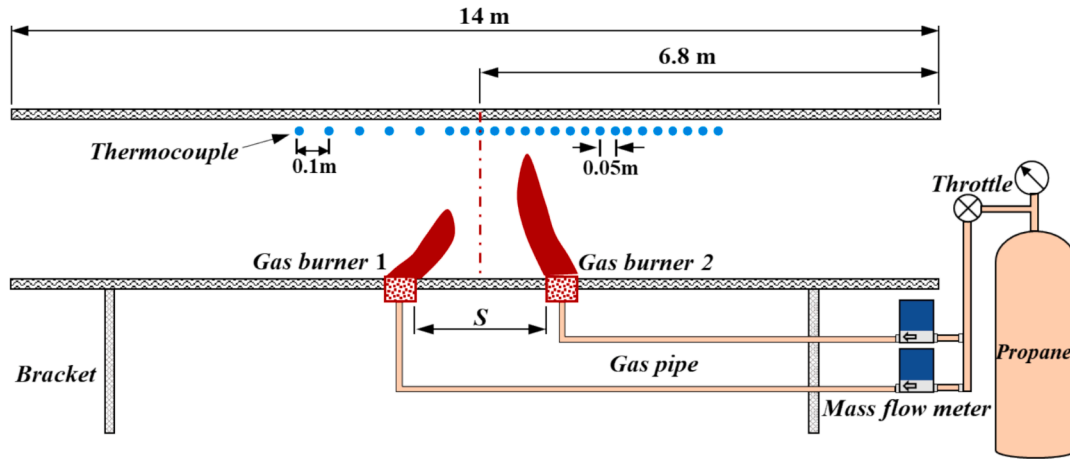


Fig. 1. Schematic diagram of the reduced-scale tunnel (front view).

Table 1

Test cases.

Test No.	Q_A (kW)	Q_B (kW)	Fire source spacing (m)
1–9	5.4	5.4	0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.7, 0.9
10–18	5.4	10.8	0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.7, 0.9
19–27	5.4	16.2	0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.7, 0.9
28–36	5.4	21.6	0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.7, 0.9

could be found: (a) with an increased distance between the two burners, the flame inclination angle first becomes greater and then approaches a steady state. (2) The flame inclination angle for the smaller fire on the left side is greater than that of the larger fire on the right side. (3) For a relatively large distance between the two burners, the flame inclination angle of the smaller fire on the left side increases with N . While the flame inclination angle of the larger fire on the right side decreases with N .

3.2. Flame height

Fig. 4 presents flame height versus fire spacing with different energy release rate ratios. The mean flames hit the ceiling directly in a few cases when the distance between the two burners is very small for few cases. In

this situation, the Z_f could be estimated using the sum of the effective tunnel height and half of the total ceiling flame length based on the previous study (Heskestad and Hamada, 1993). The same method was also used in previous studies (Gao et al., 2015; Tao et al., 2017). In addition, the position of the flame tip is estimated by ceiling gas temperature ($\Delta T_g = 550$ K) in these cases because of the shielding of the frame of the tunnel. The following phenomena can be found: (a) flame height gradually becomes lower with an increased distance between the two burners because the restricted level of air entrainment decreases and air entrained into the flame increases. For relatively long distances between the two burners, the flame interaction from the adjacent flame is limited. As a result, the induced air velocity and the flame inclination angle tend to be stable, which results in a relatively stable flame height. (b) The merged flame becomes higher with an increased energy release rate ratio N when the mean flames merge vertically. For the scenarios with two fully separated flames, the right flame generated by the larger fire becomes higher with an increasing energy release rate ratio N . While the left flame generated by the smaller fire changes a little because the energy release rate remains unchanged.

For a single fire source with axisymmetric air entrainment, an empirical model was proposed by Heskestad (Heskestad, 1983) to calculate flame height, which is expressed as:

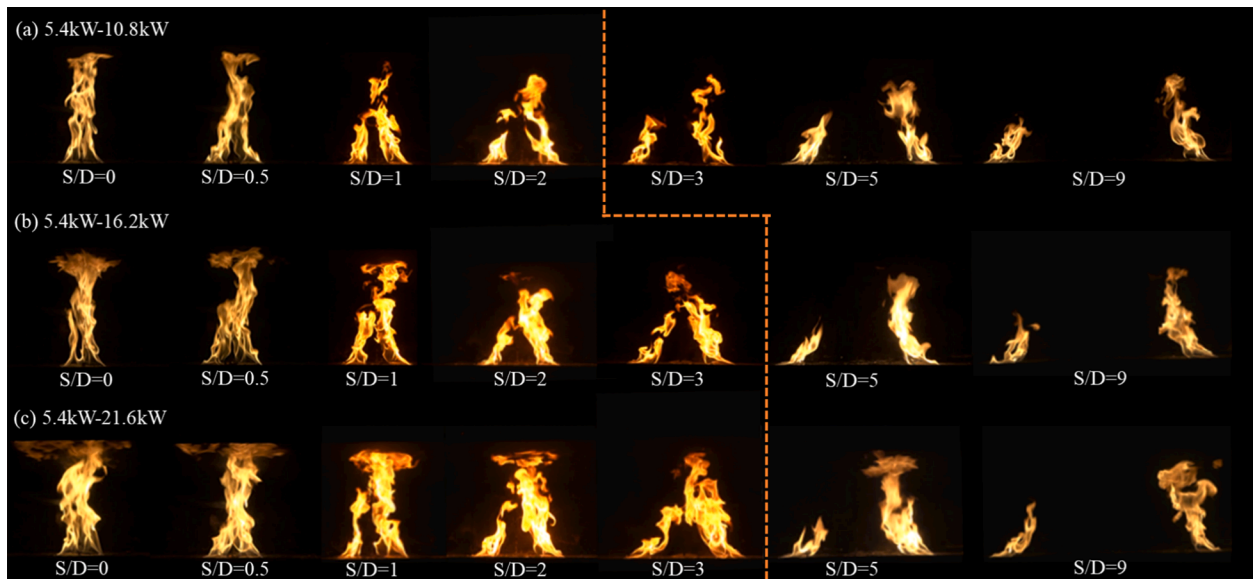


Fig. 2. Representative pictures of flame shape under different flame merging states.

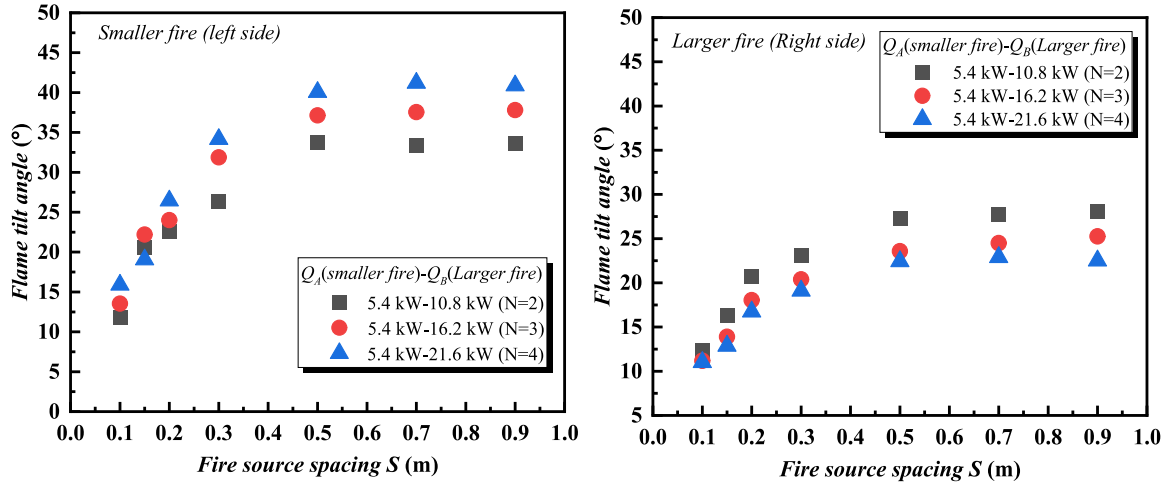


Fig. 3. The flame inclination angle versus fire spacing.

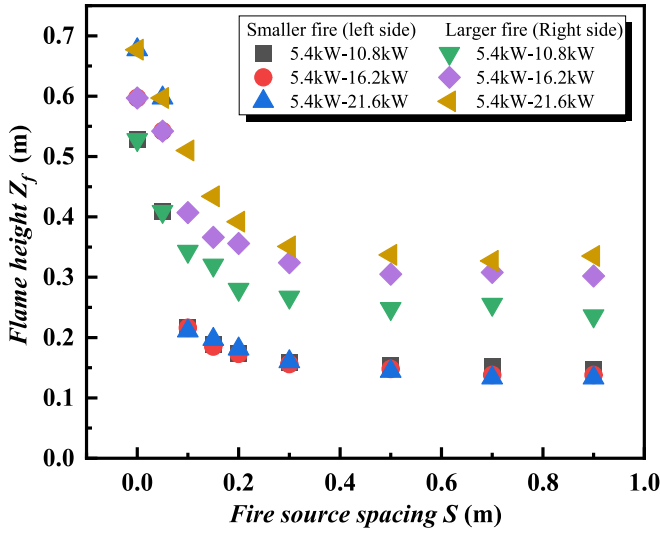


Fig. 4. The flame height versus fire source spacing.

$$\frac{Z_f}{D_{eq}} = 3.7Q_p^{2/5} - 1.02 \quad (1)$$

where D_{eq} is the equivalent diameter, and is the dimensionless energy release rate using D_{eq} as a characteristic length scale.

For mean flame merging scenarios, it can be treated as one flame. But the source of merged flame is rectangular for the two square burners. Based on the previous study (Zhang et al., 2019), the flame height is closely related to the entrainment boundary for rectangular fire sources. Therefore, the air entrainment boundary P is used to replace the equivalent diameter D_{eq} , and the following equation can be obtained:

$$\frac{Z_f}{P} = 3.7Q_p^{2/5} - 0.325 \quad (2)$$

where Q_p^* is the dimensionless energy release rate using air entrainment boundary P , which is expressed as:

$$Q_p^* = \frac{Q}{c_p T_\infty \rho_\infty g^{1/2} P^{5/2}} \quad (3)$$

When the two mean flames are merged, the total energy release rate should be used in Eq. (3). When the distance between the two burners increases from zero, the merged flame starts to separate from the bases. On one side, an additional longitudinal air entrainment boundary ($2S$)

will be caused. Meanwhile, the air entrainment side between the two burners starts to expose as the distance between the fire sources increases from zero. As a first attempt, this part of additional entrainment is assumed to be proportional to the dimensionless fire source spacing, that is, this part of additional entrainment is assumed as αD , where $\alpha = S/S_c$. When the distance between the two burners is zero, the additional air entrainment boundary is zero; when the distance between the two burners increases to the critical spacing for flame merging, the air entrainment side approaches the maximum value. Therefore, the total air entrainment perimeter for the two square fire sources could be calculated as: $P = 6D + 2\alpha D + 2S$.

Fig. 5 shows dimensionless flame height versus dimensionless energy release rate Q_p^* . The data on flame height generated by a single fire source with different burner shapes and symmetric double fire sources in previous studies (Zukoski et al., 1985; Zhang et al., 2019; He et al., 2021) are also placed in Fig. 5. It can be found that when the entrainment boundary P is taken as the characteristic length scale, flame height data with different number of fire sources cluster well together. Fig. 6 shows a comparison between the experimental flame heights and predicted flame heights estimated using Eq. (2). It clearly shows that most experimental flame height data are distributed within the $\pm 18\%$ error lines, which declares the flame height can be well predicted by Eq. (2). It

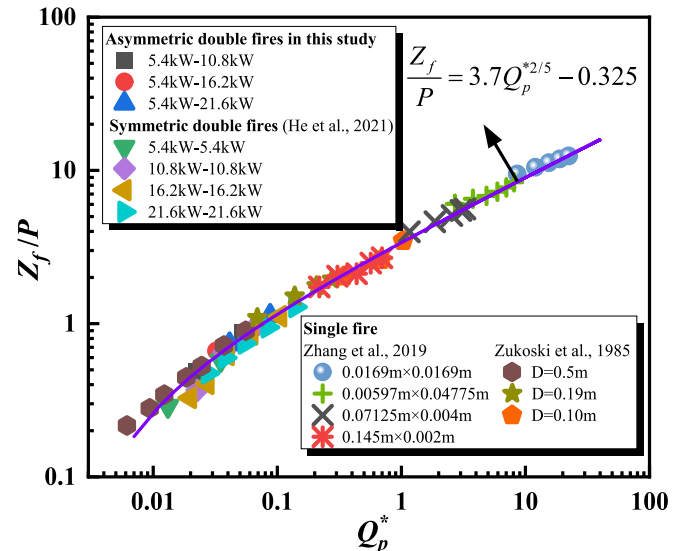


Fig. 5. Dimensionless flame height versus dimensionless energy release rate.

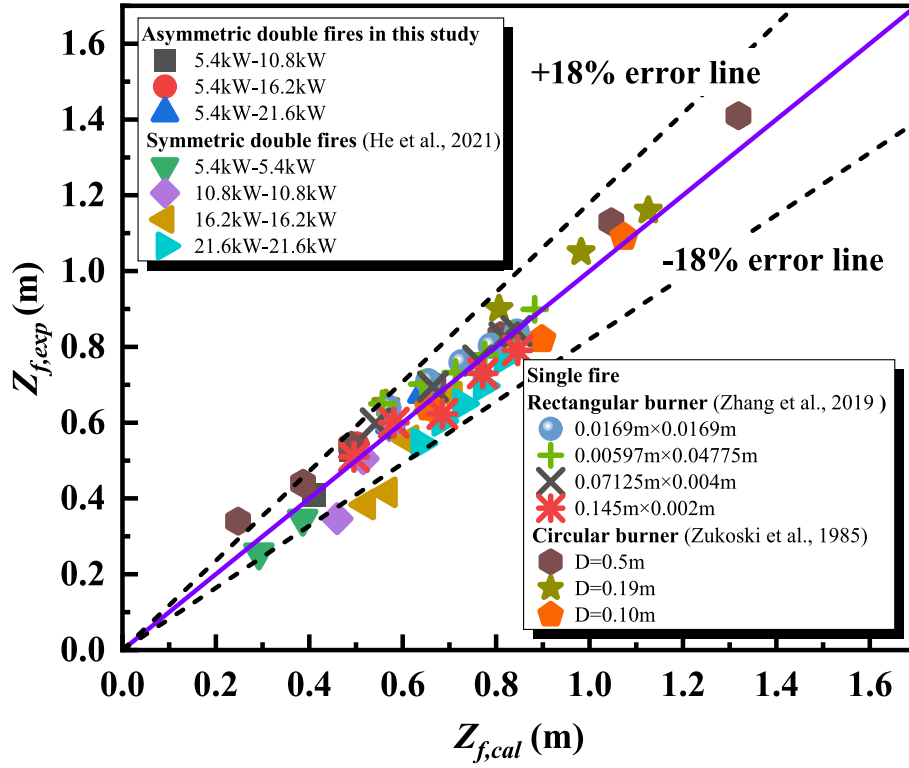


Fig. 6. Experimental data versus predictions given by Eq. (2).

should be noted that Eq. (2) is obtained under no additional wind conditions, which cannot be used to estimate the flame height for the cases using the longitudinal ventilation system.

For the cases with two separated flames, flame heights generated by left and right fire sources are disparate for the asymmetric double fires. Caused by influence from adjacent flame, flame height first becomes lower and then approaches a stable value when the distance between two burners increases. According to the study of Hu et al (Hu et al., 2017), flame interaction could be correlated to the dimensionless fire source spacing. In their study with symmetric two fires, the dimensionless fire source spacing is defined as the ratio of the distance between the centers of the two burners and the flame height of a single free flame. However, due to the different flame heights of asymmetric double fire sources, the flame height in the case with zero separation distance between the centers of the two burners is taken to replace the flame height of a single fire source. In addition, the energy release rate ratio N also has an effect on the flame properties. When the two flames are not merged, the energy release rate released by each burner also affects individual flame height. Therefore, the flame height of a free single fire with identical energy release rate is used to obtain the dimensionless flame height on both sides, respectively. Fig. 7 shows dimensionless flame height versus dimensionless fire source spacing $\dot{S}/Z_{f,s=0}$ and energy release rate ratio N . Symmetric double fire sources are only the special scenario ($N = 1$), therefore, the data of flame height of symmetric double fires are also plotted in Fig. 7. It clearly shows that experimental data under different energy release rates ratios and fire source spacings cluster well together, which can be expressed as follows:

$$\frac{Z_f}{Z_{f,s}} = \begin{cases} 0.54(SN^{0.15}/Z_{f,s=0})^{-0.39}, & SN^{0.15}/Z_{f,s=0} \leq 0.83 \\ 0.58, & SN^{0.15}/Z_{f,s=0} > 0.83 \end{cases} \quad (4)$$

Fig. 8 presents the comparison between experimental flame heights and predictions estimated using Eq. (4). It could be found that the maximum error is within $\pm 15\%$, which indicates the flame height can be well predicted by Eq. (4) in the case that mean flames cannot merge vertically. Also, Eq. (4) can be used under no additional wind conditions.

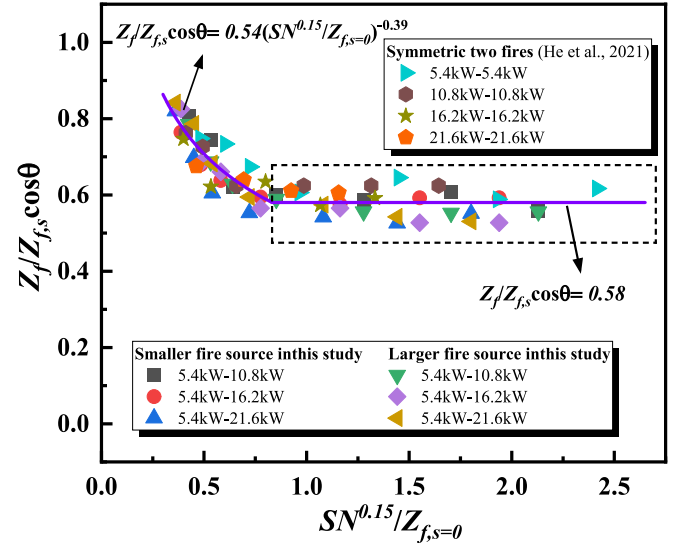


Fig. 7. Flame height versus normalized fire source spacing when the two flames are not merged.

3.3. Maximum ceiling smoke temperature

Fig. 9 presents the maximum ceiling smoke temperature versus fire source spacing. Similar to the phenomenon of flame height, the maximum ceiling smoke temperature becomes lower with an increased fire source spacing. This could be attributed to the decreased air entrainment limitation between two burners with the fire source spacing, which leads to an increased total air entrainment into the fire plume. When fire source spacing is long enough, the maximum ceiling smoke temperature gradually becomes stable. The reason is that the influence of adjacent fire does not change, resulting in the air

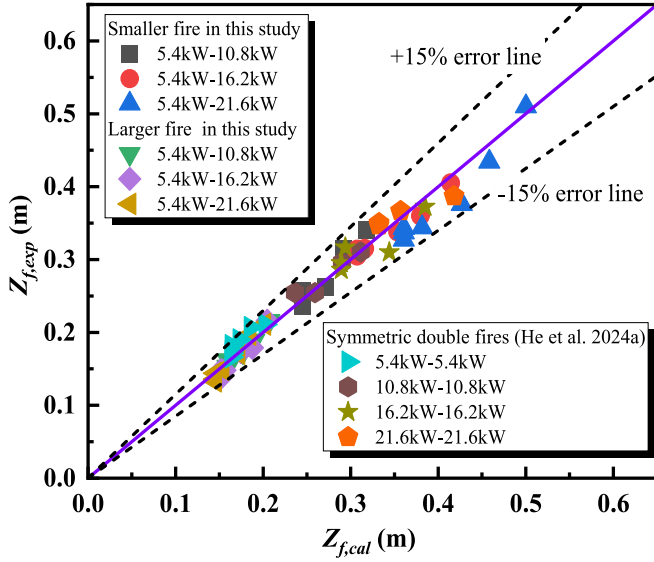


Fig. 8. Comparison between experimental results and predictions estimated using Eq. (4).

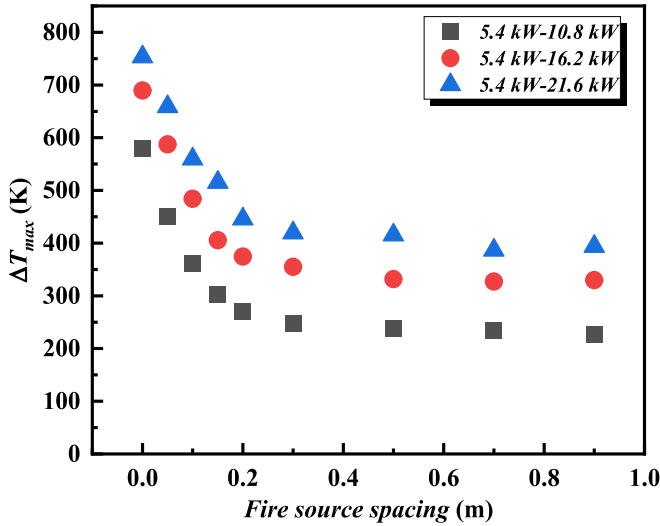


Fig. 9. Maximum ceiling smoke temperature versus fire spacing.

entrainment gradually tending to a fixed value. The maximum ceiling smoke temperature becomes higher when the energy release rate ratio N increases for a given energy release rate and fire spacing. For fire scenarios with two separated flames, this indicates that the maximum ceiling smoke temperature is mainly decided by the larger fire. When the two flames merge vertically, one merged plume hits the tunnel ceiling, as a result, the maximum ceiling smoke temperature is decided by the total energy release rate.

According to abundant experimental results in the free environment, McCaffrey divided the centerline smoke temperature into three regions: continuous flame zone, intermittent flame zone, and plume zone (McCaffrey and Flames, 1979). For fire scenarios with a restricted ceiling, three cases were divided by Gao et al. (Gao et al., 2016): continuous flame impinging on the ceiling, intermittent flame impinging on the ceiling and buoyant plume impinging on the ceiling. In their study, the index of the prediction model was consistent with that in free space. This indicates the maximum ceiling smoke temperature holds a certain correlation with the relative position of the flame tip and tunnel ceiling. Therefore, the mean flame height is used as characteristic scale length to

obtain normalized effective tunnel height, which is expressed as:

$$H_{ef}^* = H_{ef}/Z_f \quad (5)$$

where H_{ef} is the height difference between the tunnel ceiling and burner top surface. When mean flames are not merged, the large fire decides the maximum ceiling smoke temperature, as a result, the higher flame height is used in Eq. (5) for asymmetric double fires. When the mean flames merge vertically, the merged flame height is used in Eq. (5) because the total energy release rate decides the maximum ceiling smoke temperature.

Fig. 10 shows the maximum smoke temperature versus non-dimensional tunnel height (H_{ef}/Z_f). In Fig. 10, model-scale tunnel fire test data from previous studies (Ingason and Li, 2011; He et al., 2024a) with symmetric double fires and with a single fire source are also plotted. When the combustion zoom impinges on the ceiling, the maximum temperature should be unchanged as the combustion flame temperature is fixed. In the previous study (Heskestad, 1984), the excess gas temperature $\Delta T = 900 \text{ K}$ was used for continuous flame temperature, which is also adopted in this study. From Fig. 10, the maximum ceiling smoke temperature data under different fire scenarios cluster together, which can be expressed as follows:

$$\frac{\Delta T_{\max}}{T_{\infty}} = \begin{cases} 3.02, & H_{ef}/Z_f < 0.61 \\ 1.84(H_{ef}/Z_f)^{-1}, & 0.61 \leq H_{ef}/Z_f < 1.39 \\ 2.29(H_{ef}/Z_f)^{-5/3}, & 1.39 \leq H_{ef}/Z_f \end{cases} \quad (6)$$

Fig. 11 presents experimental data as a function of predictions estimated by Eq. (6). It clearly shows that experimental data are mainly distributed within $\pm 15\%$ error lines, indicating that the maximum ceiling smoke temperature with different situations can be well predicted by Eq. (6). It should be noted that Eq. (6) is proposed for the tunnel fire under natural ventilation.

4. Conclusions

The asymmetric multiple fire sources phenomenon is more common because the fire loads of different vehicles are disparate. In the present study, the flame properties and maximum ceiling gas temperature with two asymmetric fire sources in the naturally ventilated tunnel are investigated by numerous reduced-scale tunnel fire tests. The energy release rate, asymmetric ratio, and fire spacing are considered during the test. The main conceptions are summarized as follows:

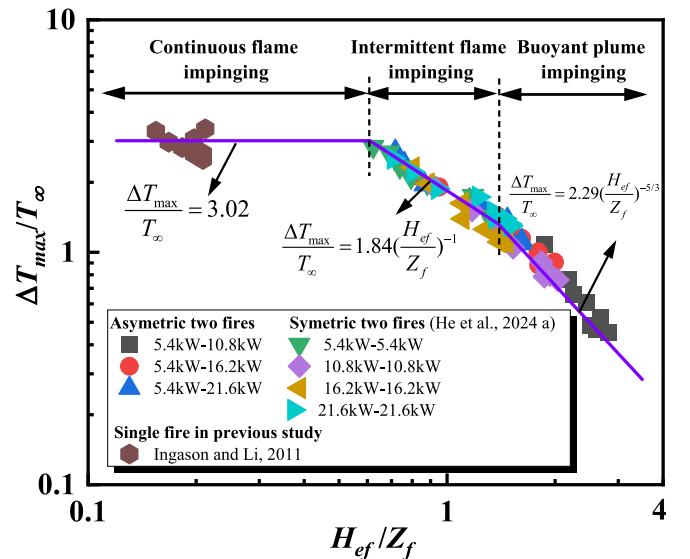


Fig. 10. The maximum ceiling smoke temperature versus dimensionless effective tunnel height.

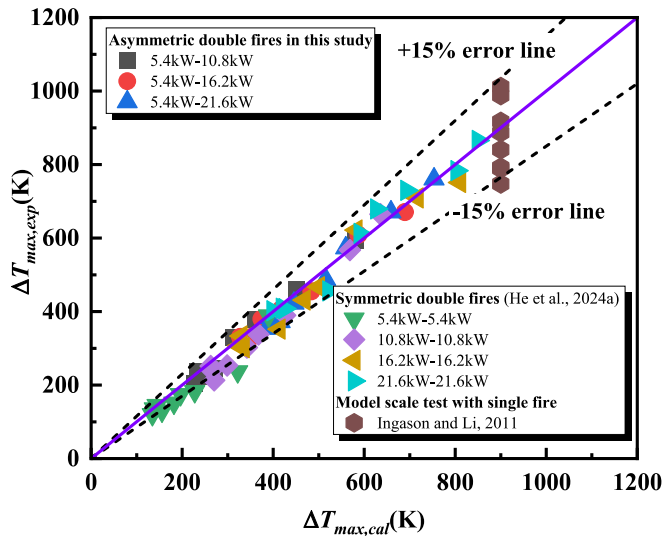


Fig. 11. Experimental data as a function of predictions estimated using Eq. (6).

- (1) with the increased distance between the two burners, the flame inclination angle first becomes greater and then approaches a stable state for the relatively long fire source spacing. For a fire source with a smaller energy release rate, the flame inclination angle is greater. Furthermore, the flame inclination angle of the smaller fire becomes greater with an increased energy release rate ratio N . In contrast, the flame inclination angle of the larger fire decreases with the energy release rate ratio N .
- (2) When mean flames merge vertically, the merged flame acts like one single fire, and the merged flame height becomes higher with an increased energy release rate ratio N for a fixed energy release rate of the smaller fire. A revised model is proposed to calculate merged flame height by taking the total air entrainment perimeter as the characteristic scale length. When mean flames cannot merge vertically, two mean flames both become lower first and then keep a steady value with a longer fire source spacing. A dimensionless correlation is also developed to estimate flame height for the cases with two separated flames.
- (3) With an increased fire source spacing, the maximum ceiling smoke temperature becomes lower first and then remains unchanged for a relatively large fire spacing. When the two flames merge vertically, the maximum ceiling smoke temperature is decided by the total energy release rate. When flames are not merged, the maximum ceiling smoke temperature is mainly decided by the larger fire. According to the relative position of the flame tip and tunnel ceiling, an empirical model for the maximum ceiling smoke temperature is developed, which is described in Eq. (6).

CRediT authorship contribution statement

Kun He: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fei Ren:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Jian Li:** Writing – review & editing, Methodology. **Xiaodong Qian:** Writing – review & editing, Methodology, Investigation. **Junyi Li:** Writing – review & editing, Investigation. **Yao Hong:** Writing – review & editing, Methodology, Formal analysis. **Wei Cong:** Writing – review & editing, Methodology, Investigation. **Yanan Hou:** Writing – review & editing, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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